

Rohan Jain

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EDUCATION

University Of Colorado, Boulder

Master's in Data Science (GPA: 3.9/4.0)

Aug 2024 - May 2026 | Boulder, USA

Mumbai University, Rajiv Gandhi Institute of Technology

Bachelor's of Engineering in Computer Engineering (GPA: 3.7/4)

Jul 2020 - May 2024 | Mumbai, India

SKILLS

Languages & Data: Python, R, SQL (MySQL, PostgreSQL; CTEs, window functions), MongoDB. **Machine Learning:** scikit-learn; supervised learning workflows, train-test splits, feature preparation, model selection, baseline comparisons, hyperparameter tuning, error analysis; evaluation (MAE, MSE, CRPS) and calibration-focused reporting. **Deep Learning:** TensorFlow, PyTorch, Keras; sequence models and custom loss experimentation with repeatable runs. **Data Quality & Reproducibility:** pandas, NumPy; schema, unit, and range validation gates; consistent artifacts, plots, and documentation for review and reruns. **Visualization & BI:** Tableau, Power BI, Excel (PivotTables, XLOOKUP, Power Query), Matplotlib, Plotly, Seaborn. **Engineering:** FastAPI, REST APIs, Git/GitHub, Docker (basics); LangChain, FAISS; Apache Spark; AWS (Lambda, S3), Azure.

EXPERIENCE

Data Science & Engineering Intern, Nexus Weather & Climate

May 2025 - Present | Boulder, CO

- Improved forecast post-processing by replacing a legacy SVM bias-correction step with a tuned Random Forest and validating results for rollout notes and production promotion; reduced temperature MAE from 0.73 to 0.39 (47%) and humidity MAE from 4.68 to 2.37 (49%).
- Strengthened uncertainty communication by building a probabilistic CNN-LSTM trained with a custom CRPS loss for calibration, then packaging simple plots and short summaries for review by non-ML stakeholders; achieved a best reported CRPS of 0.1787 for ensemble calibration.
- Standardized model comparison by creating a reproducible training and benchmarking workflow with schema, unit, and range checks, then training SVR, RF, tuned RF, GBRT, XGBoost, and MLP under the same evaluation rules; produced consistent artifacts with MAE and MSE tracked against pre-ML baselines.
- Reduced weak updates reaching later stages by adding an inference gate that requires post-ML MAE to be lower than the pre-ML baseline before a model is accepted; reduced failed runs by 90% by automatically rejecting candidates that did not beat baseline performance.
- Ran controlled loss-function experiments where only the loss changed (CRPS vs hybrid CRPS+MSE with $\alpha = 0.80$ vs Huberized CRPS) and recorded runtime plus delta CRPS outcomes; hybrid ran in 3.65 min vs 4.4 min baseline with mean delta post-CRPS of -0.01 to -0.03, while Huberized showed -0.005 to -0.02 with more stable validation behavior.

Graduate Research Assistant, Kopf Lab, University of Colorado Boulder

Dec 2025 - Present | Boulder, CO

- Improved stable isotope data quality by standardizing parsing outputs across vendor formats, enforcing consistent field names, units, and schemas so the same file yields the same structure across platforms and package versions for repeatable analysis, reporting, and downstream comparisons.
- Reduced silent data integrity issues by adding fail-fast validation for corruption and edge cases using type, bounds, and structural checks (missing blocks, wrong ordering, truncation), returning clear errors instead of partial or incorrect outputs during ingestion and analysis workflows.
- Decoded legacy Thermo Isodat scan (.scn) files stored via MFC CArchive Serialize by mapping class layouts and field order, then implementing R readers that follow the same read sequence and reconstruct analysis-ready objects with traceable provenance for debugging, review, and consistent extraction.
- Strengthened auditability by attaching file IDs, format signals, parsing paths, and assumptions to outputs, and building regression tests with golden outputs plus corruption fixtures so updates do not break prior behavior or hide parsing errors during maintenance and iteration.
- Supported secure multi-user access planning by separating authentication from app execution, using Keycloak OIDC for login and role/group access control while ShinyProxy runs one container per user session, enabling safer operations, controlled access, and maintainable updates for lab users.

PROJECTS

SaferRide (Risk-Aware Bicycle Routing) | Python, FastAPI, SQL, PostgreSQL, PostGIS,

Sep 2025 - Dec 2025

GeoPandas, OSRM, Map UI

- Built SaferRide, a routing app that requests OSRM routes, offers multiple alternatives, and scores each route using crash density and hazard proximity; cleaned city datasets, computed route-level risk with PostGIS spatial joins and GeoPandas, and served routes plus risk metrics via FastAPI with map overlays for safety versus time comparisons.

Denver International Airport Analytics | Python, Power BI, Tableau

Oct 2024

- Integrated ServiceNow and Azure DevOps data into analysis-ready pipelines, defined KPIs and built dashboards with drill-downs for SLA status, queue times, and bottlenecks; authored reusable ETL queries and KPI definitions for consistent refresh logic, reduced manual reporting by 40%, and earned 3rd place in a hackathon for dashboard impact.

AI PDF Chatbot | Python, FastAPI, React, LangChain, FAISS

Jan 2025

- Built a PDF question-answering tool for scanned and digital documents by combining OCR, embeddings, and FAISS retrieval behind a FastAPI and React UI; standardized chunking and prompt templates, supported secure uploads and real-time responses, added usage logging to track common queries, and ran labeled spot checks to refine retrieval quality.

PUBLICATION

- Technical paper: *Yoga Posture Detection and Correction*. [Link](#)